Abdullah Ejaz Janjua

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Education

Ghulam Ishaq Khan Institute of Engineering Sciences and Technology — Topi, Pakistan
Bachelor of Science in Computer Science ***(CGPA: 3.78/4.00)*** — *Sep. 2023 – June 2027*

Class Rank: 4/41 **Batch Rank:** 13/631
Dean's Honor Roll: Distinction in all semesters to date (SGPA $\geq$ 3.50).

Research Experience

LISA: Less Is Sufficient for Aging — June 2026 – Present
Mitacs Globalink Research Internship (Fully Funded), Carleton University
Advisor: *Dr. Junfeng Wen* — *Ottawa, Canada*

- Designed systematic intervention experiments on SDXL by selectively injecting source and target prompts into different diffusion blocks, identifying layers responsible for identity preservation.
- Developed a parameter-efficient LoRA-based adaptation strategy to preserve identity from a single reference image while enabling age transformations through text conditioning.
- Evaluated training-free diffusion acceleration techniques, reducing denoising steps from 50 to 8 while analyzing trade-offs in image quality and identity preservation.

DFT-DETR: Discrete Fourier Transform DEtection TRansformer for Domain Generalization for Microscopy Images — June 2025 – November 2025
Intelligent Machines Lab, Information Technology University | ***Advisor:*** *Dr. Waqas Sultani* — *Lahore, Pakistan*

- Investigated domain generalization for DETR-based object detection models on hematology microscopy images, aiming to improve performance on unseen target domains.
- Proposed a windowed Discrete Fourier Transform (DFT) approach to preserve local textural and spatial information lost through global frequency decomposition.
- Integrated frequency-domain features into intermediate model layers, improving detection performance by +5.2 mAP@50 on the LeukemiaAttri dataset.

BuddyMix: Unsupervised Domain Adaptation for Microscopic Images — Jan 2025 – Feb 2025
Intelligent Machines Lab, Information Technology University
Advisor: *Dr. Waqas Sultani* — *Lahore, Pakistan*

- Investigated uncertainty in pseudo-label generation and across-batch image mixing caused by high inter-class and foreground-background similarity.
- Proposed integrating Monte Carlo Dropout (MCD) to estimate epistemic uncertainty through stochastic forward passes, improving the reliability of pseudo-label supervision.
- Contributed to baseline experiments and manuscript preparation for a framework that improved adaptation performance by +3.4 mAP@50 on the M5 Malaria dataset.

Teaching Experience

Computer Organization and Assembly Language Lab — Jan 2026 – May 2026
Ghulam Ishaq Khan Institute of Engineering Sciences and Technology
Instructor: *Dr. Taj Khan* — *Topi, Pakistan*

- Assisted 40–50 students in the Verilog Lab by troubleshooting digital logic and HDL programming errors.
- Identified areas where students required more time and proposed schedule modifications to the instructor to ensure better comprehension.

Linux Lab Tutorial — Jan 2026 – Jan 2026
Ghulam Ishaq Khan Institute of Engineering Sciences and Technology
Instructor: *Dr. Taj Khan* — *Topi, Pakistan*

- Assisted 40–50 students in understanding Linux concepts and resolving difficulties encountered during laboratory exercises.
- Conducted technical vivas to evaluate students' understanding of Linux concepts and command-line operations.

Selected Projects

TinyTorch: Custom C++/CUDA Deep Learning Framework

- Developed a C++/CUDA deep learning framework with Python bindings (pybind11) and a dynamic DAG-based autograd engine for automatic differentiation.
- Implemented CUDA kernels for convolution, linear layers, and pooling, utilizing shared-memory tiling and memory coalescing optimizations.
- Designed a GPU-resident tensor abstraction to minimize CPU-GPU data transfers and retain intermediate tensors in VRAM during execution.
- Developed a fused Softmax-Cross Entropy CUDA kernel specialized for fewer than 64 classes, achieving 3–4× lower latency than PyTorch in targeted benchmarks.
- **GitHub:** TinyTorch

DINOv1: Self-Supervised Learning with ResNet

- Reimplemented the DINOv1 self-supervised training pipeline for ResNet, including student-teacher networks, multi-crop augmentation, and a self-distillation objective with temperature scaling and centering.
- Implemented EMA teacher updates, gradient accumulation, cosine learning-rate scheduling, and checkpoint management for model training.
- **GitHub:** DINOv1-ResNet

Generative AI: VAE & RealNVP

- Implemented Variational Autoencoder (VAE) and RealNVP architectures for probabilistic image generation, using affine coupling layers for exact density estimation in RealNVP.
- Generated and visualized image samples from both models, tracking reconstruction and KL divergence losses for VAE and negative log-likelihood for RealNVP.
- **GitHub:** VAE, RealNVP

Skills & Interests

Languages: Python, C, C++, CUDA C
Frameworks & Libraries: PyTorch, pybind11, NumPy, OpenCV
Tools: CMake, Git
Interests: Light novels, Manga/Manhwa/Manhua, Narrative-driven video games, Hiking, Jogging

Coursework

University Courses: Compiler, Algorithms, GPU Programming, Operating Systems, Linear Algebra, Linear Algebra II, Statistics, Calculus I

Self-Study: CS50x: Introduction to Computer Science (Prof. David Malan), Machine Learning Specialization (Prof. Andrew Ng), CS231n: Deep Learning for Visual Recognition (Prof. Fei-Fei Li), Computer Vision (Prof. Andreas Geiger), Diffusion Models (Prof. Stefano Ermon)